

Being honest with causal language in writing for publication

1 | INTRODUCTION

The misleading use of causal language in publication is problematic for authors, reviewers and consumers of the information. Published research in quality journals has important knowledge implications and it is, therefore, contingent on authors to use language that is accurate and appropriate to their work. Language implying unsupported causal relationships may overstate the evidence-base, especially if accepted by uncritical readers or unwitting members of the general public who may not understand how to interpret inferential statistics. The choice of language shapes academic thought across a range of areas including research, policy, clinical practice and education impacting current and future directions in the research field (Kueffer & Larson, 2014). In an age when the media vacillates on the benefits of a range of 'fads' around diet, alcohol and exercise, it can be very confusing for consumers of health services when, one day, what purports to kill you, thereafter is lauded as beneficial to health. Likewise, in the field of complementary and alternative medicines—including traditional Chinese medicine (Watson & Xue, 2020)—where the evidence-base is threadbare, spurious relationships—which are certainly not causal—are touted as evidence of cause and effect. People ought to be protected from false claims.

2 | THE NATURE OF CAUSALITY

Identifying causal relationships is a core objective of research. Establishing causation permeates the philosophy of science (Mackie, 1980), providing coherence and temporal order to our understanding of nature (Flaherty, 2011). Rothman (1995) defines a cause as: *'...an act or event or a state of nature which initiates or permits, alone or in conjunction with other causes, a sequence of events resulting in an effect'* (p. 91). In terms of health outcomes, Bonita, Beaglehole, and Kjellström (2006) identify causal factors as events, conditions, characteristics which have a role in the outcome, either alone or in combination. Both descriptions identify causal relationships involving an exposure and a resulting outcome. A cause is 'sufficient' when its presence always produces or initiates the effect and is 'necessary' if the outcome does not occur in its absence (Bonita et al., 2006). An example of a causal relationship is developing tuberculosis. While the tuberculosis bacterium is the necessary cause, there are other contributing factors such as compromised immunity and

co-morbidities which determine whether tuberculosis arises in an individual. In health research, there is almost always more than one factor contributing to the outcome of interest, making identification of causality more nuanced.

Causal inference is the process of finding an association between two factors and determining whether this is likely to be causal. Establishing a causal relationship requires not only statistical evidence but also clinical knowledge. Once an association is established, non-causal explanations including bias, chance and confounding need to be considered (Rothman & Greenland, 2005). Hill (1965) provided a set of 'considerations for causations' which provide weight to a causal interpretation for the association, namely; strength, consistency, specificity, temporality, biological gradient, plausibility, coherence, experimental evidence and analogy.

3 | STUDY DESIGN AND CAUSALITY

There is a consensus that 'temporality', which refers to the necessity that the cause must precede the effect in chronological time, is essential for establishing a causal relationship, for which longitudinal follow-up is required. Well-designed longitudinal trials can provide inferences about causation (Green, Freedman, & Gordis, 2011), while observational studies merely identify associations. Observational studies cannot provide evidence for the direction of causality, i.e. whether exposure influences the outcome, or vice versa, or whether both were influenced independently by some other confounding factors.

The concept of the 'hierarchy in the levels of evidence' in terms of validity about causation implies that well-designed randomized controlled trials (RCTs) possess the highest level of evidence (Buhse, Rahn, Bock, & Mühlhauser, 2018). Systematic reviews and meta-analyses can provide better evidence than RCTs (Evans, 2003). Systematic reviews evaluate the quality of studies and the consistency of the results across studies, while meta-analyses increase the precision of estimates by analysing the pooled data from individual studies. Among observational studies, research evidence level increases from case reports, case series, cross-sectional, case-control and cohort studies. The quality of the study design also needs to be considered. A poorly designed and conducted RCT may provide biased evidence in contrast to a well-designed cohort study (Wang & Attia, 2010). As research designs differ in terms of the risk of error and bias of their results, grading evidence is important to assess the strength of causal evidence and if the effect is worthy of reporting (Visentin, Cleary, & Hunt, 2020).

4 | LANGUAGE OF CAUSATION IN ACADEMIC WRITING

The dissemination of research is a crucial step from evidence generation to evidence consumption and decision-making of clinical practitioners, policy makers and the general public. Use of inappropriate causal language can yield inaccurate, imprecise, distorted and overstated claims (Boutron et al., 2014). Improper use of causal language such as misinterpreting correlation as causation can have potentially serious consequences, including erroneous medical decisions (Buhse et al., 2018) and contribute to misperceptions among potentially at-risk groups (Richardson et al., 2014). Several reviews demonstrate that some published studies use problematic causal language (von Elm & Egger, 2004; Glasziou & Chalmers, 2018). Describing causal relationships in academic publications plays a crucial role in how we communicate scientific research in our community and to the public (Kleinberg & Hripcsak, 2011).

Causal language involves a clause or phrase where one event, state, action or entity is explicitly presented as influencing another. There are several properties of linguistic expressions relating to causation. The use of a causative verb (e.g. increase, decrease, improve) actively in a sentence often portrays a strong causal relationship (Adams et al., 2017). Conjunctions such as 'because', 'due to' and 'since' are also used to express causal relations (Waldmann, Solstad, & Bott, 2017). Academic writers want to make their studies accessible, and interesting, which may lead them to use language that exaggerates the causality (Woloshin, Schwartz, Casella, Kennedy, & Larson, 2009). Improper use of causal language may also be inadvertently used when the authors try to vary terms throughout to minimize monotony. For example: 'social support is associated with mental well-being of older adults' becomes: 'social support improves mental well-being of older adults'. These two sentences may look similar but have different causal meaning. Writers may use cues such as 'can', 'could', 'may', 'appear to', before a verb for weaker expression or to indicate doubt (Adams et al., 2017), such as 'social support may improve mental well-being of older adults'. The choice of causal expressions could also be influenced by authors' first language (Kranich, 2011) in part due to differing causal linguistic expressions (Dunietz, Levin, & Carbonell, 2017).

Using causal language in reporting the correlational findings from observational studies is a common source of misinformation (Boutron & Ravaud, 2018; Chiu, Grundy, & Bero, 2017). Haber et al. (2018) found that 34% of studies used language too strong for causal inference, with omitting confounding variables and generalizability being the most severe issues. Rubin and Parrish (2007) found that 70% of the studies had methodological issues limiting causal inference, among which 60% used language which inflated the causal evidence. Such methodological reviews (Cofield, Corona, & Allison, 2010; von Elm & Egger, 2004; Glasziou & Chalmers, 2018) typically classify causal language issues as binary; with causation and correlation the only possibilities. Sumner et al. (2014) provide seven levels of certainty in reporting findings;

TABLE 1 Terminology of causation

Casual language	Non-causal language
• Causes • Effects, modifies • Increases/decreases • Elevates/reduces • Makes • Improves • Influences • Impacts • Results in • Induces • Effective in • Is attributable to, contributes to • Leads to • Responsible for	• Associated • Related • Correlated • Predicts • Higher • Lower • Linked to • Varies with

Note: Adapted from Adams et al. (2017); Cofield et al. (2010); Zweig and DeVoto (2015).

no mentioned relationship, statement of no relationship, statement of correlation, ambiguous statement of relationship, conditional statement of causation, statement of can and statement of causation.

Causal language should indicate how one variable affects another including the direction of causality, while non-causal language describes a relationship, irrespective of the sequence of events. Choosing appropriate language is important for reporting study results and their causal relationships. Table 1 provides a list of terminology of causation examples to describe causal and non-causal relationships.

New research builds or extends upon findings from previous research. When using causal (or non-causal) language, authors should discuss how their research fits into the existing knowledge base as well the possible sources of bias considering the study design, and errors. Including a disclaimer and/or statement of study limitations may address the risk of readers misinterpreting findings. Cofield et al. (2010) recommends being explicit in using causal language especially in titles and abstracts as these are prominently displayed. Reviewers and editors should also consider rigorous evaluation of submitted manuscripts for the use of misleading language. Readers should carefully examine the study designs and sources of bias, and whether the results are correctly reported based on assumptions of the statistical tests used (Visentin & Hunt, 2017).

5 | CONCLUSION

Scientific writing is different from general writing and the use of appropriate language is crucial. Writing for publication, by clearly communicating findings while engaging the readership, can be challenging. Given the extent of exaggerated and misleading causal language, careful attention to the language of causation is required to minimize misinterpretation, and support knowledge development. Care in being honest with causal language ensures congruence with the study design and limits inflation of evidence.

CONFLICT OF INTEREST

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AUTHOR CONTRIBUTIONS

All authors have agreed on the final version and meet at least one of the following criteria recommended by the ICMJE (<http://www.icmje.org/recommendations/>).

- substantial contributions to conception and design, acquisition of data or analysis and interpretation of data; drafting the article or revising it critically for important intellectual content.

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